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Memo on Rags for Engineering Labs
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Improvement of ETID department Labs:

As a Lab Coordinator I'm making a written discussion in hopes to build improvements in our labs in ETID. In June 2025 I shared a discussion titled "Every Engineering Lab Needs Rags" and published this as a youtube video. At that time, I shared with the department head and the season was especially busy so I took some time to put this discussion in text form.

The following text is a transcript from my Youtube video explaining our need for rags in the department. The video is 28 minutes in length, and there are minor corrections in this text for readability. To the reader: please skim through the bold text in the following pages, where I've highlighted some essential key points for our department leadership. And as time permits, you can view this video and you'll find chapters built into the playbar so you can jump to sections you find most relevant.

Figure 1 thumbnail from my video

Video Transcript

Fundamental Systems in Labs

Hi everybody, I'm David Malawey. I'm an educator for engineering like robotics and mechatronics etc. Today I'm going to talk about rags. Every laboratory in the world where we want to build things and create something from an idea, it's going to have to have rags. And I'm the type of person that wishes to have good methods passed along that start from the foundation. For a long, long time it just didn't even occur to me to talk about this. And I think the same is true for other people that do technology stuff. So I'm going to just demonstrate some methods where the lab functions based on having things that are repeatable methods, a system that flows that eliminates all the problems that occur within and during a project of many types. Those problems are suppressed or they are pre-handled by having methods in your infrastructure, your policies, your communication. This is how we move towards the best new technologies is to have the basics, the fundamental things already handled. And unfortunately in a lot of my experiences, I'm working in places where we don't have those things handled. So they have to be established. Someone has to do it even if it sounds boring, and I'm here to help.

The last video I covered topics like building 3D printed parts and gluing those together, organizing our tools and creating liners or racks and places that are modular to keep our substances. I started to talk about painting, although I haven't gone into the details. And every single one of these areas of building stuff requires sometimes solvents, sometimes simple detergents for washing things off. Today I'm going to be replacing the front panel on this barbecue smoker. It's

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automated. This is a mechatronics device with an auger turning with a motor. It's an AC motor and then it literally has a PID controller that moves the auger to deposit more fuel and meet a certain temperature. So the almost any random item that I could bring in here and show you a demonstration of, it's going to still be based on the foundation stuff. And so for example, if we're going to use acetone, this happens all the time. **Then within that process is other processes**.

Decisions Made for all Projects

And so let's just do a quick demo here. Okay. So, ABS plastic, you can read all about this or find tutorials online for performing vapor smoothing with acetone or simply just polishing the parts. And so, it performs a little bit of melting there at the surface. But this selection of substance has to have been already made and the selection of the rag has to already been made for me to just perform the task that I want and move on. Not creating more messes or creating more issues that have to be engineered with solutions. This particular substance comes from a big sheet of like 6 feet by 6 feet and it is actually diaper cloth for small fine precise jobs. We want this instead of paper towels because we don't want to leave the lint. It's lint-free. **It doesn't leave particles on the jobs that we're doing**. And also because it is pure white, this is completely unused and they don't get reused. Then **I can detect if I have the cleaning performance** and it's collecting dirt on the surface or if I've contaminated it to some degree and I need to swap it out, we get like almost an unlimited number of those. So there's 50 or so and they get sliced up. And there's a process behind every one of these things. We have a roller. It takes like less than two minutes to make 50 of those with this gorgeous Japanese steel blade. And then it's going to get sliced up on this mat. So, we're not destroying any surface and we're not dulling the blade. Okay. So, that's a built-in process that allows us to step forward to the bigger stuff.

Our Present-day Process in ETID

And so, here's my story. My technical laboratory coordinator position at Texas A&M in the ET department that started for the very first period **in 2016, and since that time we have had no rags in our facilities**. There may be one or two that were personally purchased and personally used for a project because if the organization, if the university doesn't solve this problem then the individuals have to solve this problem or we're just not going to actually do the projects. to me if you walk into any space where if it's a laboratory for building things and it doesn't have rags, they are basically telling whoever works in here “we are not going to be actually building real stuff. We are just going to have demonstrations and show and tell.” But I don't even know really where to begin except that this message has to go out because what we do is: when a student comes in and they see “obviously this is an appropriate time to use a rag,” they go to ask an administrator and they say, "Well, we solved this problem. You can **go take a journey to the bathroom, de-glove, demask whatever it is that you've got on safety wise, and go yank rolls and rolls of sheets and sheets out of the bathroom dispenser** for paper towels". So, the paper towel is not the right thing for the job, and it's not the right process to leave and go in and out. And by the way, we have an access card, and so your hands are going to be full with something that may end up spilling. You're going to have contaminants maybe on your hands and with the stuff and just **none of this is actually an acceptable repeatable process**.

I'm going to show you the process here in our lab in this I'm calling it OpenLab or the SCUTTLE Robotics lab. We have two bins. Okay. So, this is where you will always be able to find these regular cotton white (used but clean) fresh rags. So that depending if it's a project like I'm going to be wiping this down, I can grab one that was used a little bit so far yesterday. Once it's contaminated, then it goes into that bin. And then they're going to stack up. And long before this is completely ~~open~~ [full] is the time that this one is up to here and we run it through the laundry. We have a system for doing this and it's all right here. **But with a bigger institution there are better systems.** So that's going to go into the laundry unit into the drying unit and **we won't be waiting for those things to be finished**. We will [still have clean rags] during the whole time.

Technology in rags

So this other this is also pure cotton and **rags are a technology**. If you go on to grainger.com or an industrial supplier, you will find rags like this one has a tiny amount of lint that may come off on a project. And this one you can see the

knitting style. Okay**, this is going to leave next to zero contaminants**. Um, and so this is it's more rigid. You don't include any... the process planning goes deep. It's like you would not ever want to have a waxy substance in this when you're washing it. It's pure detergent and you're not going to include something like if you use fabric softener that will cause there to be some water resistance to this. Suddenly, it's not absorbing nearly as well. And it can further deposit that, it'll smear the waxy, very, very small amount of fabric softener stuff back onto the stuff. And then when you need to, let's say, apply some epoxy here, then that epoxy is not going to stick. And so, everything starts with clean components or components that are in the condition that you actually expect. All of this should be repeatable. And **if the organization is a hundred years old, then the processes to have that fundamental stuff ought to be 90 years old at least**.

Scaled Solution for Organizations

Here's the simple solution. Um, this goes back to my days in a fraternity in undergrad and Sigma Nu fraternity Gamma chapter at Rolla, Missouri. And we had a large facility... the soaps that go into our soap dispensers and other supplies in general on a routine there. This was every single week. They're going to go and collect your rags that we use to clean the kitchen and the facility in general. and then they're also give you the fresh ones and they don't collect 100% and they're not counting them. So if you put wear and tear on the rags, if you lose some of them or whatever, then the replacements come. There's no counting up of the inventory that we're burdened with. And the company does that and they just supply the same amount every week that they supply them. And it relieves the staff and the faculty or relieves the users of this place from having to count that. But we do still have to have the process.
And so when you walk into a place, then we're going to need to know where is some cleaning substance if we're working with oils. For example, we have drilling and tapping processes here that are going to leave metal flakes or metal chips. And the process to handle that is already worked out. So, we're going to say drill or tap something here. And there will be metal flakes around this area. There's within arms reach. We have the vacu to clean up to suck up the parts that are not saturated in oil. And a brush to basically get it out of that area. Okay. Now, there's only wet particles remaining on our surfaces. And that is the time when you ought to have a rag. This rag is going to be used for the duration of the product and project. It'll pick up these little bits and it's going to stay nearby. Maybe it'll hang right here, but we're not going to need to pull out a new one every time. There could be 50 occasions of drilling and tapping and back and forth. That rag stays right here. It accumulates the small bits and there's going to be an oil very similar to the consistency of motor oil. And that's getting deposited on the holes where we're drilling. That keeps the tools cool, etc. And then some of that is actually going to get into our brush for cleaning. Now, if the brush gets saturated, we're bringing the rag over and we're going to soak it up. Now compare how easy this is to having to make the decision how we're going to handle this or can we hide it enough that the next user doesn't have to find it. That's worst case. Best case is now every time I leave a bit of oil, I don't know what's the expectation of me on how clean this ought to be. The signal of how I ought to leave the place comes from how it was before and what equipment is made available to me.

The Burden on Students

So all the issues of needing students to help out more and clean up the projects comes from the infrastructure and the processes that are just obvious. As soon as you're here, if you look up here, ooh, the safety glasses are hanging. And look at these safety glasses. When I'm at A&M, most labs don't have safety glasses. In our building, we require students to bring their own. If they forget, it's going to be a whole trip home. But the ones that are supplied, they are scratched up in every possible way. They're not maintained or replaced on a regular basis. That is in the mechanical engineering building that I recall in my during my master's degree. We could basically put on these safety goggles that make it very difficult for you to see. It's not good enough for a world-class institution. Okay, I'll come back to the solution side. , we had, I think it was Cisco Foods that also brought and returned the equipment that we use that are semi-consumable like rags at Sigma Nu Fraternity and for some monthly fee or included in the overall distribution of products to and fro is the cost of the rags, the cost of the laundry. So that we are not asking. So in the general space at the departments I've been in in Texas A&M is okay if you want to do this right you paper towel is not going to cut it, then the person's going to bring their own equipment, track their own equipment and then you have an engineer who should be having his mind dedicated to the engineering task just piling up with other things to consider. “Oh have I brought that equipment? Do I have a liner for my backpack to keep the dirty rag so I can bring it home?” And now we have engineers who **are trained**

and expected to be doing engineering. **Instead, they are doing laundry for the university**. That is the reality of where we are right now.

Our Institution's Priorities

And so one of the things like I just saw this video about Texas A&M building an artificial lunar surface for us to do research and so forth, but we have to have the fundamentals. So, $200 million is getting allocated to build this new facility where we'll have a simulated moon. And I wonder where are they going to borrow the processes that get us the very very basics to be satisfied? Are they going to have rags at that facility or are we going to have more people just burdening the staff and the students with "just figure it out on your own"? **It is as if we took technology several hundred years back to before there were already established solutions** for that.

Example Solutions Continued

So once again you pay the same distributor that's bringing supplies in your facility to also bring you rags and then they're going to give you maybe a crate or a dispenser. And in our case, we have dispensers arranged. If it's good enough, it stays forever for the next 100 years. If it's not good enough, it gets changed. But the budgeting, so like dispenser for these nitrile gloves and we have dispensers for soap, right? This stuff is all so very obvious, but it must exist there. So, if you want to wash your hands, then there ought to be a degreaser. So, this is an orange clean degreasing soap. This goes together with the rag. So, sometimes you're going to need to spread some of this substance on the rag to do the job of cleaning a surface, and it has to already be there. That's how we become a hundred times more efficient is we solve all the problems that actually exist when people build things.

Success of Students

The students who appear to be thriving in our department are actually outsourcing their lab experience.

I am far far too aware that the students that really excel and make magnificent projects... one of them I spoke with and he has his uncle's house in Houston. So he would take his break where okay it's the middle of the semester they're working on a project he's going to collect all the parts of the project at home into his car drive down to Houston and **during the break he puts in the real effort to actually get something built because he knows he can't do it in our facility at A&M**. And by the way, this is kind of true for many many other organizations it's not just A&M. But think about applied engineering, anywhere where that's happening and the academicians are present and there's research happening. If they're telling you that we're doing actual applied research, then you're going to see these signals. Are there rags? Are there soaps? Is there a place to wash something off, get hot water? **You're going to see those things. If you don't see those things, then you're being misled about what actually happens in that space**.

So what tends to happen, we have that student that had a wonderful amazing one in a million opportunity to use his uncle's workshop in Houston. If he gets it all done in that [recess] then you have the imbalance of team efforts for the project outcomes. You do not have a fair playing field. In one essence, the student with the most resources has the most privilege and the other side the student with the most resources is going to do all the work. And chances are, he also has instruction there at his uncle's workshop as well. He can get tips and be safe and figure out the processes that he ought to be learning at school. And that all those students in the team of four or five: they ought to have access to learning that stuff.

After Graduation

Those that impacts the confidence of the student in such a massive way. When they go into a job interview and they know they have actually done something and they know that their project that they worked on in theory, the things that they studied in books actually they have physically themselves applied that to a real world thing and produced an outcome that works the way that the theory works, **then they feel much more powerful**. The way that they feel if they don't have that opportunity is that they really have no idea that they could be a straight A student, but **they have no idea whether their grades matter in the real world.** And it will impact everything down to the way that they speak in

the interview to get their first job. It will completely change if you have the experience, the hands-on experience. You go into a job interview, several questions aim to extract what is the essence of that person and their interests. Now, they have a story to tell. They cannot say the story wrong because you know from your project experience this is what happened. This is what I had access to. Oh, you did something the wrong way? “Well, that was the limitation of what was available to me.” They are not stumbling through telling that story.

And I personally have interviewed engineers that have hands-on experience and ones that don't. The ones that don't are just as smart, but they don't even know what their skill set is. They don't know which of their grade A studies is worth talking about. Because they have never been able to connect the two sides of the theory and the physical. And so with a grade alone you can gauge yourself according to your peers, but with the hands-on you can gauge yourself as every engineer should with the results that happen in real life. You no longer need to compare yourself to your peers. And so this is all stuff that we are robbing from the students by not having the fundamental things in the labs.

Burden on the Lab Staff

And then the burden should not be on the lab staff to figure out which... we don't have the privilege or the budget etc to choose a supplier who's going to arrange this stuff. We if we don't have facility access then we can't just modify as I can here build a shelf and say okay that's where we're going to put the supplies. It's safe and the person has access through that door. I know that the students that's going to need the supply over here also has access to that door. It gets far too much to be on the lab staff to handle. This is a facility level and **not just department level but college level solution that has to be implemented.**

Rags, Vapors, Processes

So back to the acetone situation, if we're doing a whole project, then this contaminated rag that has some vapors coming off of it. Now, we can take a bag out of our dispenser. This can be taught with just a small pictogram on the wall of how do we handle these things? And the communication and the methods all happen fairly autonomously once they're actually good choices that can go in here. And when we close this off with a rubber band or something, now the vapors are not going everywhere. They're not going in our lungs, etc. And it's not reasonable to say that if you use one little squirt of one little chemical, then this has to be done with a hood because we don't have hoods in most of the labs. Can't... and we can't go outdoors unless you want to be in the blinding sun and 120 degrees heat of Texas. So now there's just a little bit of vapors. That's the time when we come down here and we're going to turn on here. I might as well just show you. We're going to turn on this Honeywell filter. It's a tap. This is usually on all the time. And then we can increase it from low to medium or high. And for just a few minutes, we're going to run this. And the carbon filter, the pre filter, which can be replaced. And we have extras of it. This carbon pre filter that all these black pre filters that you find in units like this Honeywell... they capture VOCs. And so is it 100%? No. But it is far far far better than having nothing or in an unexpected situation, you emit a little bit of smoke from something or you have a spill or a container that was supposed to be sealed that had a leak, then we can immediately turn this on. And now we're cycling the air in here. And you'll notice with just a little bit of vapors, the individual who's working may not smell it after two minutes. Your nose kind of adjusts if it's just a little. And then we can run this for a few minutes, invite somebody in from the outside and say, "Hey, do you smell this? Do you smell acetone? Do you smell paint thinner?" "No, we don't." "Okay." And so you can fairly well trust that this is not... I'm not creating safety standards here in this instruction. I'm saying we have ways to improve it better from having no solution at all.

Other ingredients that go together with this... almost any glue for doing a glue project, we have the silicone mats available. We have not just one but more than we need so that you can grab another one off of the shelf, lay it down, and then you're going to do a gluing project. You're not expecting to spill any on the table, but then if you do, you can roll that up and you can go let this cure on the shelf or outside and the glue will come right off when it's done. So, silicone is part of our standard processes even if it's not laid out. Um, then we also have like just hard panels. These come from flooring tiles, but there has to be some solution instead of no solution. If you do something that has rust or dust or things coming off of it and you want to be able to just pick up your mess and then carry it over and slide that into

the trash can that we we must have existing methods for all this. Okay. And then this can get wiped off. The vinyl tiles, they're rigid enough to stay flat and carry with one hand. They're resistant to almost as many glues as the silicone stuff doesn't stick to them very much. And if they do, like this looks like one of the urethane adhesives and we can use the mineral spirits to take it off.

So on the fly **you are not asking the engineer to make decisions about every subprocess within the process**. You offer those. **This is the whole purpose of having a 100-year-old university** or with Toyota, the reason we had standards for more than 50 years and we translated things all the way from Japanese over to our offices in Detroit. We do that because then the burden is not on the young new engineers to solve 100-year-old problems. That's just the truth for you. So yeah, we can take off the urethane, whatever it is. All right. So mineral spirits tends to dissolve other sticky substances and then you can come back with alcohol. By the way, if I'm using any more than just a little bit, then I glove up for that stuff. Gloves are important. Um, acetone, isopropyl alcohol is less aggressive on the skin and isopropyl alcohol will do the taking away that residual glossy stuff that you can still feel on your fingers. You can still smell it from the mineral spirits. Mineral spirits is kind of like mineral oil. It's the thinner the thinnest element of that petroleum stuff. And but it's still basically like an oil with I don't know 50% VOCs. We run over this with alcohol and that this rag absorbs the remaining mineral spirits and just a little bit of time later like 1 minute this will be completely dry and clean. Then we take the project and we set it over here on the table. Create a gap. Let the air circulate. Okay. Now the whole world doesn't have to smell it. And the vapors can get sucked up and then this is ready to go back in the bin.

Concluding

So this is what it looks like when you have an organization that wants you to succeed at building things. We create value in the institution that is covering all of your bases. So when you go there and you want to become a world-class engineer or researcher or something then you are dedicating your time to that [engineering] activities and not solving the problem of what do I put my project on. Okay that's a wrap. Now you know that every lab where things are built there ought to be rags, and if there are no rags things are not actually getting built. Thanks everybody. Have a great day.